

Exercise 1:

Define two new functions, *function1* and *function2* that accept an integer as an argument. Call them one after the other from *ASM_Function*, making sure that the callee returns every time to the caller.

- *function1* should increase the value of the argument by 5
- *function2* should decrease it by 2

Exercise 2: Compare numbers

Write an ARM assembly program that compares two **signed** integers and branches based on the result. If the first integer is greater than the second, the program should set the result to 1. Otherwise, it should set the result to 0.

Exercise 3: Loop with Branching

Write a program that sums all integers from 1 to a given number $n > 1$. Implement the loop using branching instructions.

Exercise 4: Nested Branches (Maximum of Three Numbers)

Write an ARM assembly program that finds the maximum of three unsigned integers. The numbers are passed in registers *r0*, *r1*, and *r2*, and the result (maximum value) is returned in *r0*.

Exercise 5: if-else control flow

Write a program that takes an integer as an input and returns 1 if the input is positive, -1 if the input is negative and 0 if the input is zero

Bonus: exercise 6: Write a program that computes the n -th Fibonacci number